METRIC
MIL-DTL-24728/3A
18 September 2018
SUPERSEDING
MIL-I-24728/3
1 June 1995

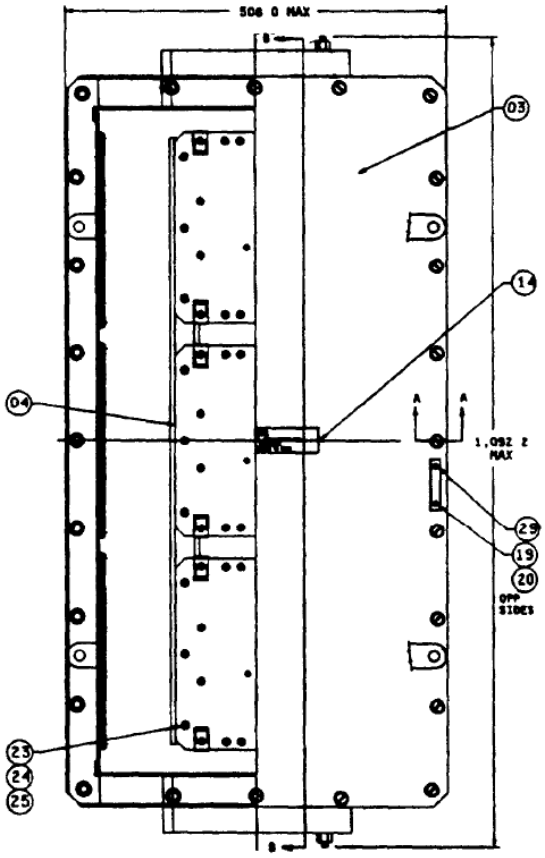
DETAIL SPECIFICATION SHEET

INTERCONNECTION BOX FIBER OPTIC, SUBMERSIBLE, 406.4 X 863.6 MM

Reactivated after 18 September 2018 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24728.



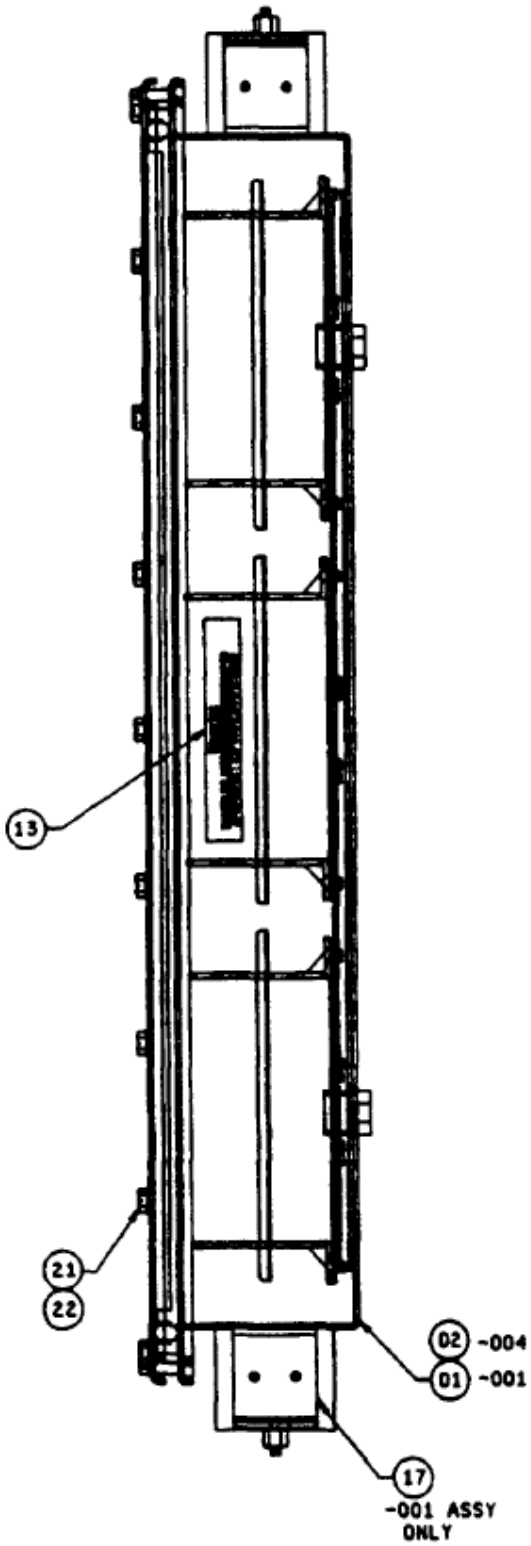
ABOVE COVER SHOWN PARTIALLY REMOVED

FIGURE 1. Enclosure assembly, M24728/3-001 and M24728/3-004.

AMSC N/A FSC 6099

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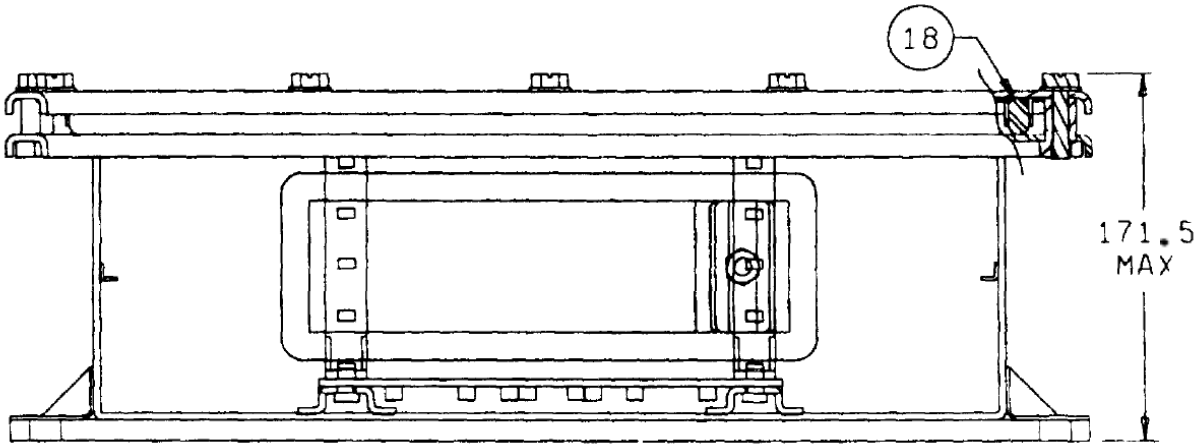
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SECTION B-B

FIGURE 1. Enclosure assembly, M24728/3-001 and M24728/3-004 – Continued.

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SECTION A-A

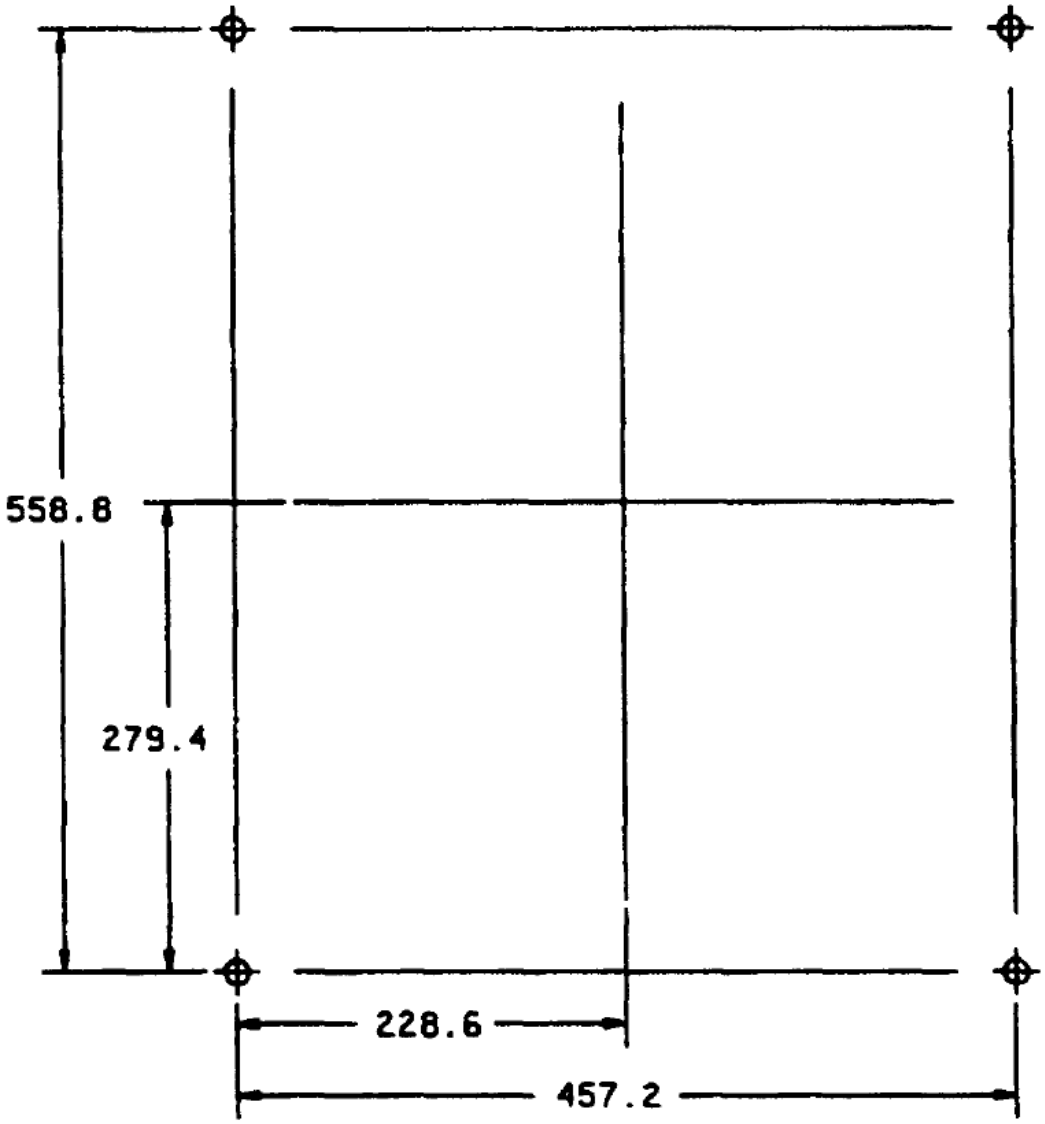
Millimeters	Inches
171.5	6.75
508.0	20.00
1092.2	43.00

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.
3. Tolerance is ±2.5 mm (0.1 inch) for one-place decimals and ±0.3 mm (0.01 inch) for two-place decimals.
4. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 1. Enclosure assembly, M24728/3-001 and M24728/3-004 – Continued.

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Millimeters	Inches
12.7	0.50
228.6	9.00
279.4	11.00
457.2	18.00
558.8	22.00

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.
3. Tolerance is ±0.3 mm (0.01 inch) for two-place decimals.
4. Mounting holes shall accommodate M12.7 x 1.95 (0.50-13) bolts.

FIGURE 2. Box mounting dimensions.

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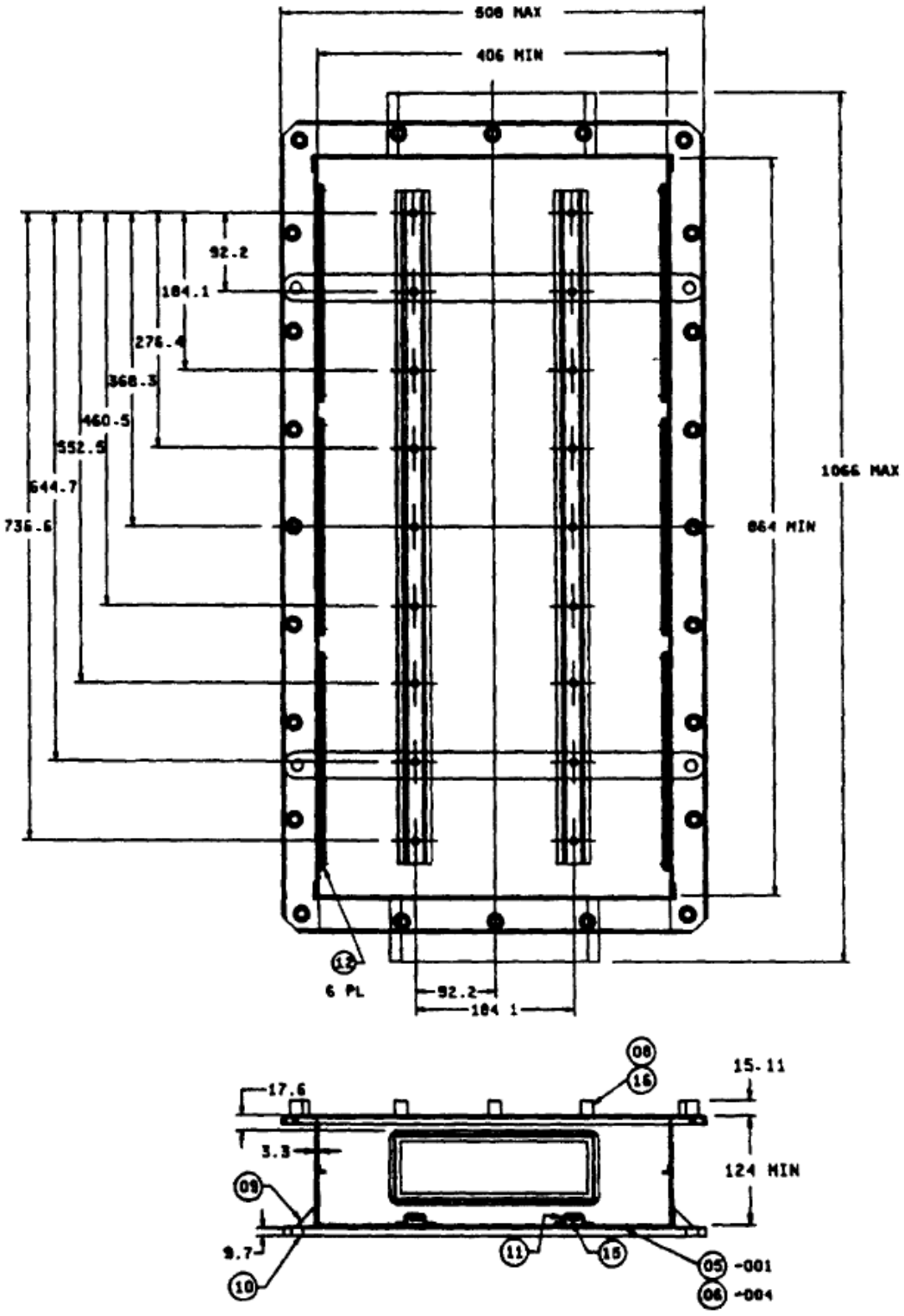
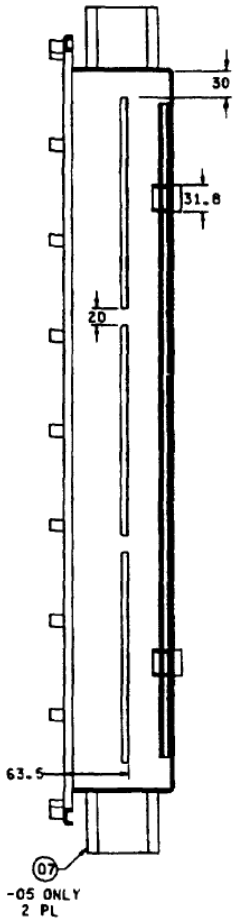


FIGURE 3. Box assembly.

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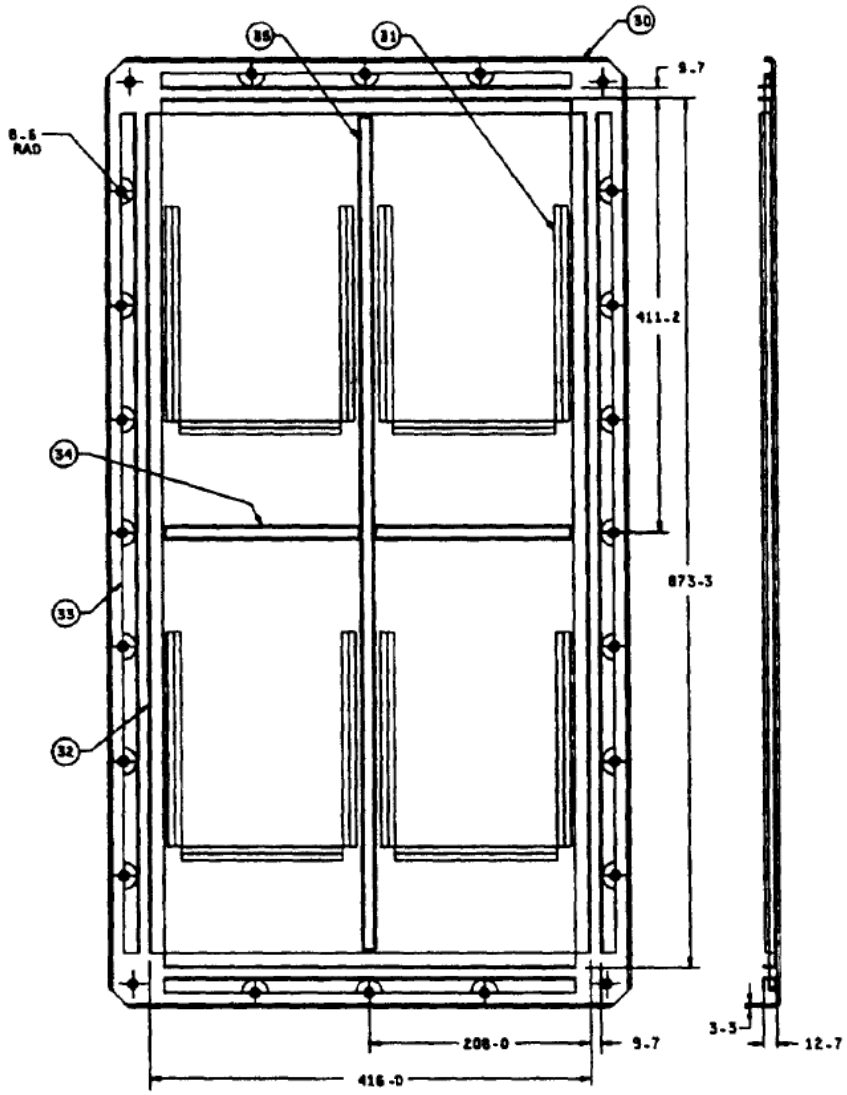
Millimeters	Inches	Millimeters	Inches	Millimeters	Inches
3.3	0.13	63.5	2.50	460.5	18.13
9.7	0.38	92.2	3.63	508	20.0
15.11	0.595	124	4.9	552.5	21.75
17.6	0.69	184.1	7.25	644.7	25.38
20	0.8	276.4	10.88	736.6	29.00
30	1.2	368.3	14.50	864	34.0
31.8	1.25	406	16.0	1066	42.0

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.
3. Tolerance is ±2.5 mm (0.01 inch) for one-place decimals, ±0.3 mm (0.01 inch) for two-place decimals, and ±0.13 mm (0.005 inch) for three-place decimals. A tolerance of ±0.8 mm (0.03 inch) is acceptable on dimensions controlled by welding or brazing, provided that this tolerance does not interfere with the interchangeability of assemblies or parts.
4. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 3. Box assembly – Continued.

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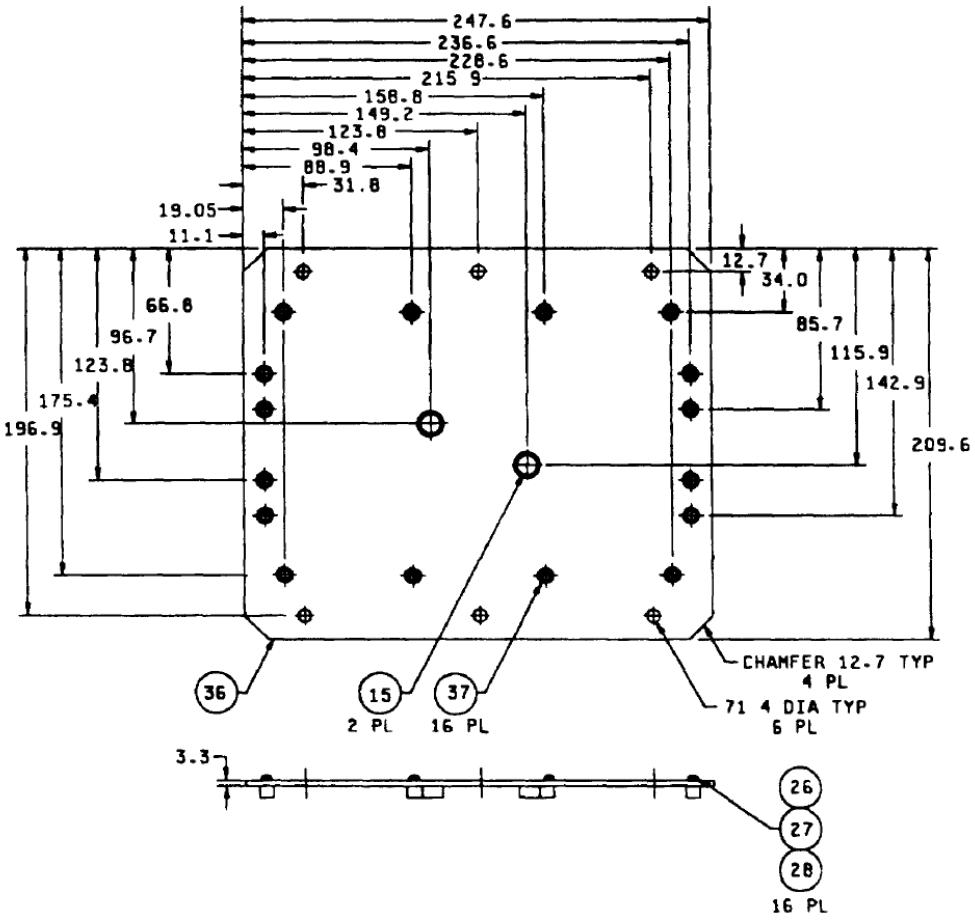
Millimeters	Inches	Millimeters	Inches
3.3	0.13	208.0	8.19
8.6	0.34	411.2	16.19
9.7	0.38	416.0	16.38
12.7	0.50	822.5	32.38

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.
3. Tolerance is ±0.3 mm (0.01 inch) for two-place decimals.
4. Detailed layout and placement of diagram holders is not specified.
5. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 4. Cover assembly.

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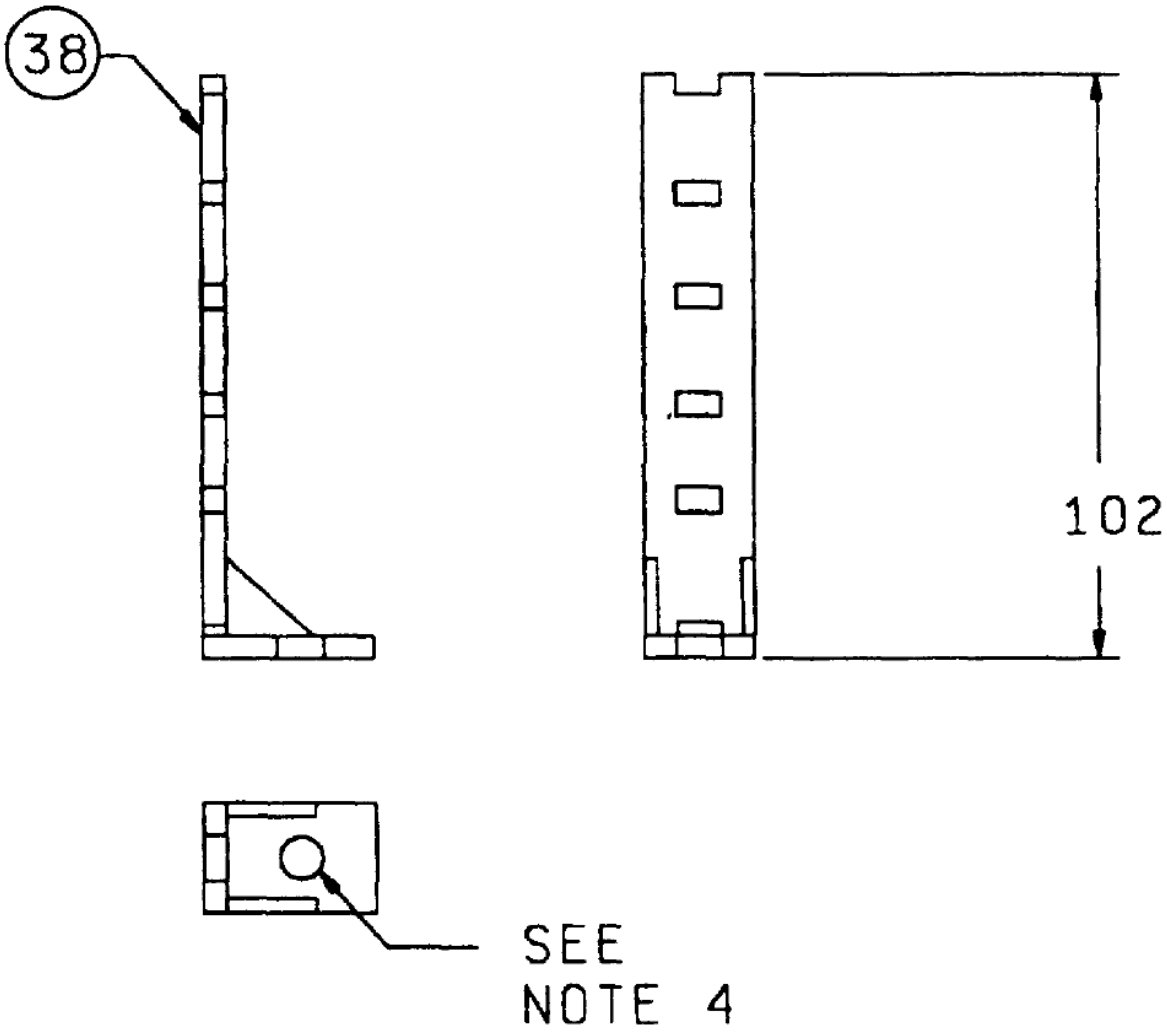
Millimeters	Inches	Millimeters	Inches	Millimeters	Inches
3.3	0.13	66.8	2.63	149.2	5.88
7.1	0.28	85.7	3.38	158.8	6.25
11.1	0.44	88.9	3.50	175.4	6.91
12.7	0.50	96.7	3.69	196.9	7.75
19.05	0.75	98.4	3.88	209.6	8.25
31.8	1.25	115.9	4.56	228.6	9.00
34.0	1.34	123.8	4.88	236.6	9.31
64.1	2.53	142.9	5.63	247.6	9.75

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.
3. Tolerance is ±0.3 mm (0.01 inch) for two-place decimals and ±0.13 mm (0.005 inch) for three-place decimals.
4. Tie wrap post is shown on figure 6.
5. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 5. Mounting plate assembly.

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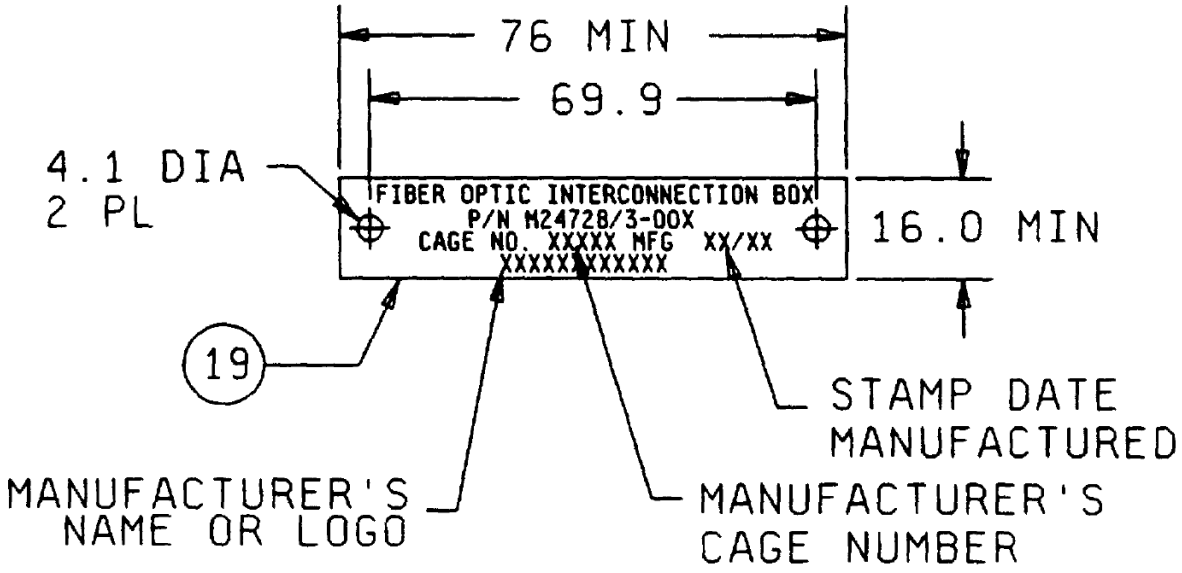
Millimeters	Inches
102	4.0

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound are given for information only.
3. Tolerance is ±2.5 mm (0.1 inch) for one-place decimals.
4. Hole is 5.6 mm (0.22 inch) diameter.
5. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 6. Tie wrap post.

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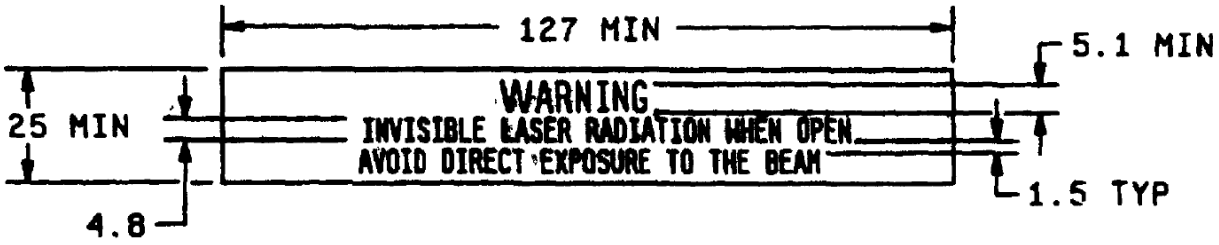
Millimeters	Inches
4.1	0.16
16.0	0.63
69.9	2.75
76	3.0

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.
3. Tolerance is ±2.5 mm (0.1 inch) for one-place decimals and ±0.3 mm (0.01 inch) for two-place decimals.
4. For identification of subcomponent parts (circled numbers), see table I.

FIGURE 7. Identification plate.

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Millimeters	Inches
1.5	0.06
4.8	0.19
5.1	0.20
25	1.0
127	5.0

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.
3. The signal word “WARNING” shall be in yellow on a black background in the upper panel. The lower panel shall consist of black letters on a yellow background.
4. Tolerance is ±2.5 mm (0.1 inch) for one-place decimals and ±0.3 mm (0.01 inch) for two-place decimals.

FIGURE 8. Inside warning label.

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Millimeters	Inches
1.5	0.06
3.3	0.13
5.1	0.20
25	1.0
127	5.0

NOTES:

1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.
3. The signal word “WARNING” shall be in yellow on a black background in the upper panel. The lower panel shall consist of black letters on a yellow background.
4. Tolerance is ±2.5 mm (0.1 inch) for one-place decimals and ±0.3 mm (0.01 inch) for two-place decimals.

FIGURE 9. Outside warning label.

TABLE I. Parts list.

Subcomponent identification numbers, circled (see figures 1 through 9)	Part name	PIN M24728/3 dash number		Subassembly number				Remarks and reference figures
		-001	-004	01	02	03	04	
01	Box assembly with MCT frame	1		1				1 and 3
02	Box assembly without MCT frame		1		1			1 and 3
03	Cover assembly	1	1			1		1 and 4
04	Mounting plate assembly	3	3				1	5
05 1/ 5/	406.4 x 863.6 box with MCT frame			1				3
06 1/ 5/	406.4 x 863.6 box without MCT frame				1			3
07 1/ 2/	MCT frame			2				3
08 1/ 6/	Flange pad			24	24			3
09 7/ 8/	Gusset			4	4			3
10 1/	Mounting leg			2	2			3

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TABLE I. Parts list – Continued.

Subcomponent identification numbers, circled (see figures 1 through 9)	Part name	PIN M24728/3 dash number		Subassembly number				Remarks and reference figures
		-001	-004	01	02	03	04	
11 1/	Mounting channel			2	2			3
12 1/	Tie wrap bracket			6	6			3
13 3/	Warning label (inside)	1	1					1 and 8
14 3/	Warning label (outside)	1	1					1 and 9
15 4/	M6.35 x 1.27 (0.25-20) insert			18	18		2	3 and 5
16 4/	M9.53 x 1.59 (0.375-16) helicoil			24	24			3
17 2/	MCT wedge pack	2	2					1
18 9/	Gasket	1	1					1
19 12/	Identification plate	1	1					1 and 7
20 12/	Blank nameplate	1	1					1
21 4/ 13/	M9.53 x 1.59 (0.375-16) hex bolt	24	24					1
22 4/	9.53 (0.375) flat washer	24	24					1
23 4/	M6.35 x 1.27 (0.25-20) x 15.9 PHMS	18	18					1
24 4/	6.35 (0.25) flat washer	18	18					1
25 4/	6.35 (0.25) split lock washer	18	18					1
26 4/	M4.0 x 79 (8.32) x 12.7 PHMS						12	5
27 4/	4.0 (#8) flat washer						12	5
28 4/	4.0 (#8) split lock washer						12	5
29 4/	M3.5 x 79 (6.32) x 6.35 PHMS	4	4					1
30 1/ 5/	406.4 x 863.6 cover	1	1					4
31 5/ 10/ 12/	Diagram holder					4		4
32 11/	Inner gasket retainer					4		4
33 11/	Outer gasket retainer					4		4
34 11/	Stiffener					1		4
35 11/	Stiffener					2		4
36 12/	Mounting plate						1	5
37 4/	M4.0 x 0.79 (8-32) insert						16	5
38	Tie wrap post	12	12					5

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TABLE I. Parts list – Continued.

NOTES:

1/ Aluminum, 5052 H-32 or 6061 in accordance with ASTM B209.
2/ See MIL-DTL-24705/1 for guidance.
3/ In accordance with MIL-DTL-15024, type G, adhesive-backed, metal foil.
4/ Corrosion-resistant steel (CRES).
5/ See MIL-E-24142 for guidance.
6/ See MIL-E-24142 for guidance.
7/ Aluminum, 5052 H-32 or 6061 in accordance with ASTM B209, 2.39 mm (0.094 inch) thick.
8/ See MIL-E-24142for guidance.
9/ See MIL-E-24142 for guidance.
10/ Constructed to hold a 152-mm (6-inch) by 279-mm (11.0-inch) diagram.
11/ Aluminum, 5052 H-32 or 6061 in accordance with ASTM B209, 2 mm (0.060 inch) thick (min.).
12/ Aluminum, 5052 H-32 or 6061 in accordance with ASTM B209.
13/ See MIL-E-24142 for guidance.

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REQUIREMENTS:

Design and construction. See figures 1 through 9 and table I.

Interconnection box. The interconnection box shall be constructed using MIL-E-24142 as guidance.

MCT. The Multiple Cable Transit (MCT) shall be constructed using MIL-DTL-24705/1 as guidance. The MCT interior dimensions shall be 217.4 by 60.0 millimeters (8.56 by 2.36 inches).

Enclosure assembly construction. The enclosure assembly shall be constructed using the following combinations of materials:

a. The box, MCT frame, flange pads, gussets, mounting legs, mounting channels, tie wrap bracket, and cover shall be aluminum, 6061 in accordance with ASTM B209.

b. The box, MCT frame, flange pads, gussets, mounting legs, mounting channels, tie wrap bracket, and cover shall be aluminum, 5052 H-32 in accordance with ASTM B209.

Enclosure assemblies constructed using aluminum, 6061 shall not be welded.

Enclosure assemblies constructed using aluminum, 6061 shall be dip-brazed in accordance with AWS C3.7.

Enclosure assemblies constructed using aluminum, 6061 shall be artificially aged at 177 °C for 12 hours to a 6061-T4 condition.

Enclosure assemblies shall be class 1A conversion coated in accordance with MIL-DTL-5541 before being epoxy powder coated.

Cable interconnection interface. The cable feed-through shall be in accordance with MIL-DTL-24705 and shall accept cables having outer diameters between 6 and 25 millimeters (0.24 and 1.00 inch) constructed in accordance with MIL-PRF-85045.

Enclosure assembly stress relief. Enclosure assemblies shall be stress relieved in accordance with MIL-E-24142.

Mass M24728/3-001 20.4 kilograms (45 pounds maximum)
M24728/3-004 18.2 kilograms (40 pounds maximum)

Interconnection box mounting. See figure 2.

Interconnect organization.

Fiber optic diagram holders. See figure 4.

Fiber optic cable tie post. 12 (see figure 6).

Fiber optic splice tray holder. Three maximum (see MIL-DTL-24728/8).

Fiber optic connector module. Three maximum (see MIL-I-24728/6).

Fiber optic splices. When outfitted with MIL-DTL-24728/8 splice tray holder modules, the interconnection box shall organize up to 48 MIL-PRF-24623/6 fiber optic splices per splice module (maximum of 3 splice modules).

Fiber optic connector adapters. When outfitted with MIL-I-24728/6 connector modules, the interconnection box shall organize up to 48 MIL-DTL-83522/17 connector adapters (maximum of 3 connector modules).

Cable retention. Not applicable.

Cable seal flexing. Not applicable.

Cable twist. Not applicable.

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Compression resistance. Not applicable.

Impact. Not applicable.

Water pressure. The applied pressure shall be 44.8 kilopascals (6.5 pounds per square inch), equivalent to a depth of 4.57 meters (15 feet). The interconnection cover bolts and the MCT wedge pack nut shall be tightened to a minimum torque of 16.94 newton-meters (150 inch-pounds) during this test.

Flammability. Not applicable.

Salt spray. Not applicable.

Flame spread. Not applicable.

Fungus test. The fungus test shall be conducted only on polymeric materials.

Operating and storage temperature range. The operating temperature range shall be -54 to +65 °C and the storage temperature range shall be -62 to +85 °C.

Part or identifying number (PIN). See MIL-DTL-24728 and table I.

M24728/3-001, box, mounting plates, and lid with cable entrance and wedge pack.

M24728/3-004, box, mounting plates, and lid without cable entrance.

Gasket size. The gasket size is 415.9 millimeters (16.38 inches) wide by 873.1 millimeters (34.38 inches) long.

Insert block. Insert blocks compatible with the MCT frame are specified in MIL-DTL-24705/1.

Measurement system. The U.S. Government preferred system of measurement is the metric (SI) system. However, this item was originally designed using inch-pound units of measurement. In the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85

Preparing activity:
Navy – SH
(Project 6099-2017-004)

Review activities:
Army – AR, MI
Navy – AS, CG, EC, MC, YD
Air Force – 19, 99
DLA – CC, GS
NASA – NA

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